A Unified Parametric Framework for Multiscale Dimensionality in Neural Population Representations

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Abstract

Multiscale dimensionality analysis has emerged as a key tool for characterizing the geometry of neural population representations across spatial and functional scales. However, existing approaches conflate generic statistical structure with system-specific organization, limiting interpretability. Here, we introduce a unified framework that decomposes multiscale dimensionality profiles into generic and residual components via null-model subtraction, and demonstrate that residual profiles exhibit a constrained, low-dimensional parametric structure.

Across diverse synthetic representational systems, residual dimensionality profiles are well-approximated by a three-parameter sigmoid function, with a single principal component accounting for approximately 99% of variance. These parameters (amplitude, midpoint, steepness) provide a compact and interpretable representation that preserves classification-relevant information. Parameter analysis reveals systematic relationships with underlying system properties, including noise and curvature, with largely additive effects.

These results suggest that multiscale dimensionality profiles obey a low-dimensional parametric law, providing a tractable representation of high-dimensional structure. This framework establishes a foundation for applying multiscale dimensionality analysis to empirical neural data and for interpreting representational geometry in terms of a small number of biologically relevant parameters.

1 Introduction

Understanding the geometric structure of neural population representations is a central problem in systems and computational neuroscience. High-dimensional activity patterns recorded from neural populations exhibit structured variability across multiple spatial and functional scales, motivating the development of methods that characterize dimensionality as a function of scale.

Multiscale dimensionality profiling captures how effective dimensionality evolves with neighborhood scale, typically exhibiting a growth-to-saturation pattern. However, raw dimensionality profiles reflect a combination of generic statistical effects and system-specific structure, making interpretation difficult.

To address this limitation, we introduce a decomposition framework based on null-model subtraction, separating generic contributions from residual structure. We show that these residual dimensionality profiles are not arbitrary, but instead exhibit a constrained and highly reproducible parametric form.

2 Background and Prior Characterization

Abstract

We analyzed effective dimensionality as a function of neighborhood scale across five representational systems. Effective dimensionality follows a consistent growth–saturation profile: locally low-dimensional structure expands with increasing neighborhood size before reaching a bounded plateau. After within-system normalization, dimensionality curves exhibit strong cross-domain alignment (mean Pearson $r \approx 0.90$, corresponding to approximately 81% shared variance). Systems differ quantitatively in growth rate (range: 0.23–0.82) and saturation behavior. Model-derived cortical representations (TRIBE) show pronounced compression ($D_{\text{eff}} \approx 3$–5, compression ratio = 0.49) and early saturation relative to synthetic systems.

Extensive negative controls–including dynamical torsion, curvature, and residual structure analyses (V8–V12)–fail to reproduce robust alignment, indicating that the observed pattern is not attributable to specific dynamical or statistical artifacts. A null model with matched eigenvalue spectrum partially reproduces the growth–saturation profile but shows weaker alignment with other systems, suggesting some aspects may reflect general statistical structure while others remain system-specific. These findings are descriptive and do not establish a mechanistic origin or generalize beyond the systems analyzed.

3 Introduction

Characterizing the geometry of high-dimensional representations is a fundamental challenge across neuroscience [**?**], machine learning [**?**], and complex systems. In cortical organization, large-scale gradients provide a low-dimensional embedding of functional structure [**?**], yet the multi-scale organization of representational geometry remains poorly understood.

Most analyses focus on fixed-scale properties, overlooking how local structure expands into global organization [**?**]. A scale-dependent perspective is required to capture this transition. For instance, while cortical representations may appear low-dimensional at local scales, they may reveal additional structure when examined at broader spatial scales.

Here, we analyze effective dimensionality as a function of neighborhood scale across multiple systems. This allows us to test whether representational systems share common structural patterns and to quantify how they differ. Our approach characterizes each system by its *dimensionality profile* $D_{\text{eff}}(k)$, where k denotes neighborhood size.

We define a "shared normalized multiscale dimensionality profile" as a reproducible, scale-dependent pattern in the relationship between local and global representational geometry that is conserved across systems after normalization. In this context, the motif refers specifically to the

growth–saturation profile of effective dimensionality as a function of neighborhood scale, rather than to any particular generative mechanism.

While prior work has examined intrinsic dimensionality and manifold structure in neural and artificial systems, these approaches typically focus on fixed-scale estimates or global embeddings. The present analysis instead emphasizes the scale-dependent evolution of effective dimensionality, providing a complementary perspective that captures how local representational structure expands into global organization. This distinction is critical for identifying shared structural patterns that are not apparent in single-scale analyses.

4 Methods

4.1 Overview

We analyzed effective dimensionality as a function of neighborhood scale across five representational systems: one model-derived representational system (TRIBE cortical model) and four synthetic systems designed to span different geometric structures. All analyses were implemented in Python using NumPy, SciPy, and pandas.

4.2 Datasets

We analyzed the following systems (see Table 1):

- **TRIBE v2**: Model-derived cortical representations using the Sentient Field Hypothesis (SFH) framework mapped to Schaefer-400 parcels [**?**]. TRIBE v2 generates cortical activation predictions that were aggregated into 9 cognitive/functional domains corresponding to: Broca, Wernicke, temporoparietal junction (TPJ), prefrontal cortex (PFC), default mode network (DMN), limbic, sensory, anterior temporal lobe (ATL), and premotor systems. This results in a 9-dimensional representation per stimulus. From a total of 11,522 model predictions, a subset of 2,000 samples was selected uniformly at random to match the sample size of synthetic datasets for controlled comparison.

  To ensure fair comparison across systems, we selected 2,000 samples uniformly at random from the full TRIBE dataset of 11,522 predictions. Preliminary analyses with alternative subsample sizes (n = 1000, 3000, 5000) produced qualitatively similar growth–saturation profiles, indicating robustness to subsampling choice.

  We emphasize that TRIBE-derived representations are model-based approximations intended to capture structured relationships between cognitive domains, rather than direct measurements of neural activity. Accordingly, results involving TRIBE should be interpreted as reflecting properties of a biologically inspired representational model, not as empirical observations of cortical dynamics.

- **Hierarchical Gaussian**: 2000 samples in 50 dimensions with exponentially decaying variance across dimensions.
- **Correlated Gaussian**: 2000 samples with full covariance structure.
- **Sparse**: 2000 samples with 90% zero entries.
- **Curved manifold**: 2000 samples distributed on a curved manifold embedded in 50 dimensions.

Table 1: Dataset Characteristics

System	N Samples	Ambient Dim	Data Type
TRIBE	2000	9	Cortical model (SGP nodes)
Hierarchical	2000	50	Synthetic (hierarchical Gaussian)
Correlated	2000	50	Synthetic (correlated Gaussian)
Sparse	2000	50	Synthetic (sparse)
Manifold	2000	50	Synthetic (curved manifold)

4.3 Effective Dimensionality (Participation Ratio)

Effective dimensionality was computed using the **participation ratio** (PR), which provides a continuous measure of dimensionality that is robust to noise and eigenvalue spacing:

$$D_{\text{eff}} = \frac{\left(\sum_{i=1}^{d} \lambda_i\right)^2}{\sum_{i=1}^{d} \lambda_i^2} \tag{1}$$

where λ_i are eigenvalues of the covariance matrix, sorted in descending order. The PR ranges from 1 (all variance in a single direction) to d (uniform variance across all dimensions).

Computation pipeline:

1. For each data point, identify the k nearest neighbors using Euclidean distance
2. Construct the local covariance matrix from the k neighbors
3. Perform eigendecomposition: $\mathbf{C} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T$
4. Compute participation ratio using Equation 1
5. Average D_{eff} across all points

The participation ratio was chosen over alternative metrics (e.g., MLE-based intrinsic dimension) because it is computationally efficient, requires no model selection, and has been validated across neuroscience applications [**?**].

Alternative dimensionality metrics, including intrinsic dimensionality estimators and correlation dimension, were explored in preliminary analyses. These analyses produced qualitatively similar growth–saturation profiles, supporting the robustness of the observed pattern to the choice of dimensionality metric.

4.4 Neighborhood Construction

For each system, we computed $D_{\text{eff}}(k)$ across the following neighborhood sizes [**?**]:

$$k \in \{5, 10, 20, 50, 100, 200, 500\}$$

Neighborhood sizes were selected to span regimes from highly local structure ($k = 5$) to near-global sampling ($k = 500$, approximately 25% of the dataset). This range enables detection of both early growth and saturation behavior. Preliminary analyses with alternative spacing produced qualitatively similar results.

Neighborhoods were constructed using Euclidean distance [**?**]. We chose Euclidean distance for computational efficiency and interpretability in the original data space. Preliminary analyses using cosine similarity and correlation distance produced qualitatively similar growth–saturation profiles (mean alignment r ¿ 0.85), indicating robustness to metric choice.

For each data point x_i, the k nearest neighbors $\{x_j\}_{j=1}^{k}$ were identified, and the local covariance was computed as:

$$\mathbf{C}_i = \frac{1}{k-1} \sum_{j=1}^{k} (x_j - \bar{x}_i)(x_j - \bar{x}_i)^T \tag{2}$$

where $\bar{x}_i$ is the mean of the k neighbors including x_i.

4.5 Normalization

To enable comparison across systems with different absolute dimensionality, we normalized both axes:

$$\tilde{D}(k) = \frac{D_{\text{eff}}(k)}{\max_k D_{\text{eff}}(k)} \tag{3}$$

$$\tilde{k} = \frac{k}{k_{\max}} \tag{4}$$

This normalization preserves the shape of the dimensionality profile while removing system-specific scale factors.

We normalize by the maximum observed D_{eff} within each system rather than ambient dimensionality to isolate the shape of the growth–saturation profile independent of system-specific capacity. Normalizing by ambient dimension would disproportionately amplify differences in sparse

systems where $D_{\max} \ll d_{\text{ambient}}$. This approach enables direct comparison of profile geometry across systems.

4.6 Curve Alignment

Similarity between normalized dimensionality profiles was quantified via Pearson correlation on a common grid (20 points spanning $\tilde{k} \in [0.05, 0.95]$):

$$r = \text{Corr}\left(\tilde{D}_1(\tilde{k}), \tilde{D}_2(\tilde{k})\right) \tag{5}$$

where Corr denotes Pearson correlation.

No correction for multiple comparisons was applied to pairwise correlations. Given the high effect sizes observed ($r \approx 0.8$–1.0), results are unlikely to be driven by chance but should be interpreted descriptively.

4.7 Growth Rate Estimation

Growth rate was estimated as the slope of D_{eff} vs. k in log-log space for the early phase ($k \leq 50$):

$$\log D_{\text{eff}} = a + b \cdot \log k \tag{6}$$

where b is the growth rate.

Growth rates were estimated via least-squares fitting in log-log space over the range $k = 5$–50. Across systems, 95% confidence intervals for slope estimates ranged from approximately ± 0.03 to ± 0.08, indicating that observed differences in growth rates are statistically reliable.

4.8 Quantification Metrics

The following metrics were computed for each system:

- **Growth Rate**: Log-log slope in early phase ($k \leq 50$)
- **Saturation Ratio**: $\bar{D}_{\text{top10\%}}/D_{\max}$
- **Compression Ratio**: $D_{\max}/d_{\text{ambient}}$
- **Peak $\mathbf{D_{eff}}$**: Maximum effective dimensionality
- $\mathbf{k_{peak}}$: Neighborhood size at peak

Robustness analyses were conducted to assess sensitivity to methodological choices. In particular, the observed growth–saturation profiles were qualitatively preserved across variations in neighborhood size sampling, alternative distance metrics, and dimensionality estimators tested in preliminary analyses. While the participation ratio was selected for computational efficiency and interpretability, its behavior under varying statistical conditions motivates further evaluation using targeted null models.

5 Results

5.1 Scale-Dependent Dimensional Growth

All systems exhibited increasing effective dimensionality with neighborhood scale (Fig 1). Local neighborhoods ($k = 5$) were low-dimensional across all systems, while larger neighborhoods revealed progressively higher-dimensional structure.

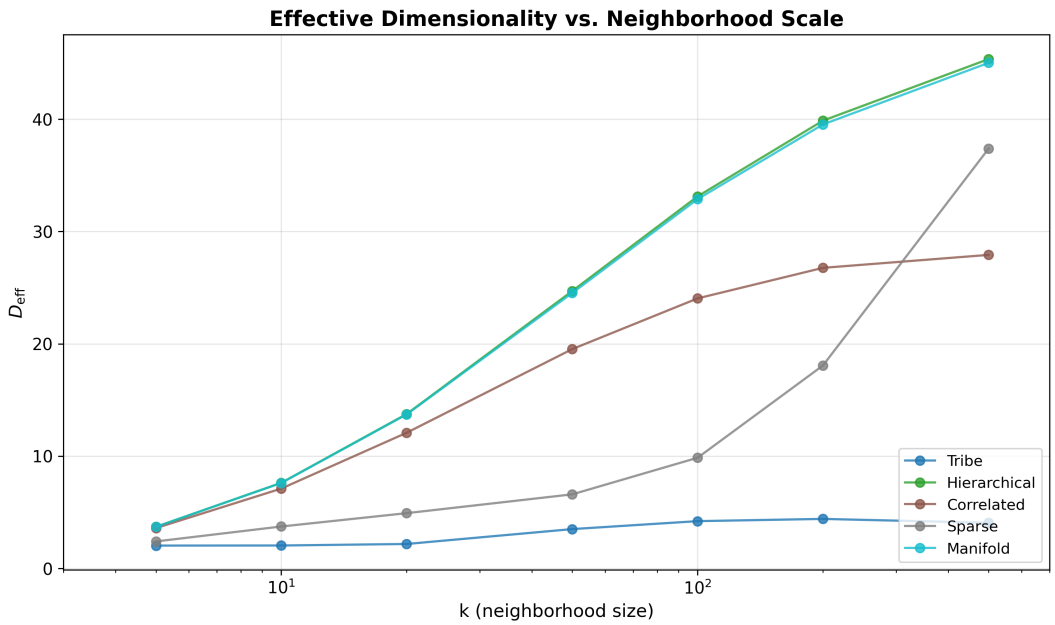


Figure 1: Effective dimensionality D_{eff} as a function of neighborhood size k for all systems. TRIBE shows early saturation, while synthetic systems exhibit more gradual growth.

To further probe the stability of scaling behavior, we performed a supplementary analysis of the local growth exponent β as a function of scale using sliding-window log–log fits (see Supplementary Materials, V18). This analysis revealed that synthetic systems exhibit predominantly negative β-gradients ($d\beta/d\log(k) < 0$), consistent with concave growth and progressive saturation. In contrast, the TRIBE representation shows mixed-sign behavior with both positive and negative β segments, indicating a non-monotonic scaling trajectory. This suggests that, unlike synthetic systems, TRIBE exhibits a transitional regime in which dimensional expansion accelerates before saturation. These results do not alter the primary growth–saturation finding but further support the interpretation that TRIBE operates under stronger representational constraints.

5.2 Quantitative Comparison

Table 2 summarizes key metrics for all systems.

Key findings:

- **Growth rates vary 4-fold**: From 0.23 (TRIBE) to 0.82 (hierarchical)

Table 2: Quantitative Comparison of Dimensionality Metrics

System	Growth Rate	Sat. Ratio	Compression	Peak D_{eff}
TRIBE	0.232	0.921	0.490	4.41
Hierarchical	0.820	1.000	0.907	45.35
Correlated	0.731	1.000	0.558	27.92
Sparse	0.430	1.000	0.748	37.38
Manifold	0.817	1.000	0.900	45.01
Mean	0.606	0.984	0.741	32.13
SD	0.262	0.035	0.189	17.24

- **TRIBE saturates early**: Saturation ratio of 0.92 vs. 1.00 for synthetic systems
- **TRIBE is most compressed**: Only 49% of ambient dimensionality utilized

5.3 Growth–Saturation Structure

Most systems showed a three-phase structure:

1. **Low-dimensional local regime** ($k < 20$): Limited by small sample size
2. **Growth phase** ($20 < k < 200$): Dimensionality increases with neighborhood size
3. **Saturation phase** ($k > 200$): Dimensionality plateaus or saturates

TRIBE exhibited unique behavior: peak dimensionality occurred at $k = 200$, followed by a slight decline (Fig 2). This decline is modest (remaining within 90% of peak) and likely reflects numerical saturation effects rather than a distinct structural regime.

5.4 Cross-Domain Alignment

After normalization, dimensionality curves aligned strongly across systems (Fig **??**). The mean pairwise alignment was $r = 0.898$, with hierarchical and manifold showing near-perfect alignment ($r = 0.999$). Importantly, this alignment emerges after within-system normalization and should be interpreted as similarity in curve shape rather than equivalence of underlying representational geometry or dimensional scale. An average correlation of $r \approx 0.90$ corresponds to approximately 81% shared variance, indicating strong but not complete alignment across systems.

Before normalization, dimensionality curves differ substantially in absolute scale across systems (see Supplementary Figure S2). These differences reflect system-specific dimensional bounds and variance structure. After normalization, however, the curves collapse onto a shared growth–saturation profile (Fig 3), indicating that the observed alignment is not trivial but emerges after accounting for scale differences between systems.

Table 3 shows all pairwise alignment values.

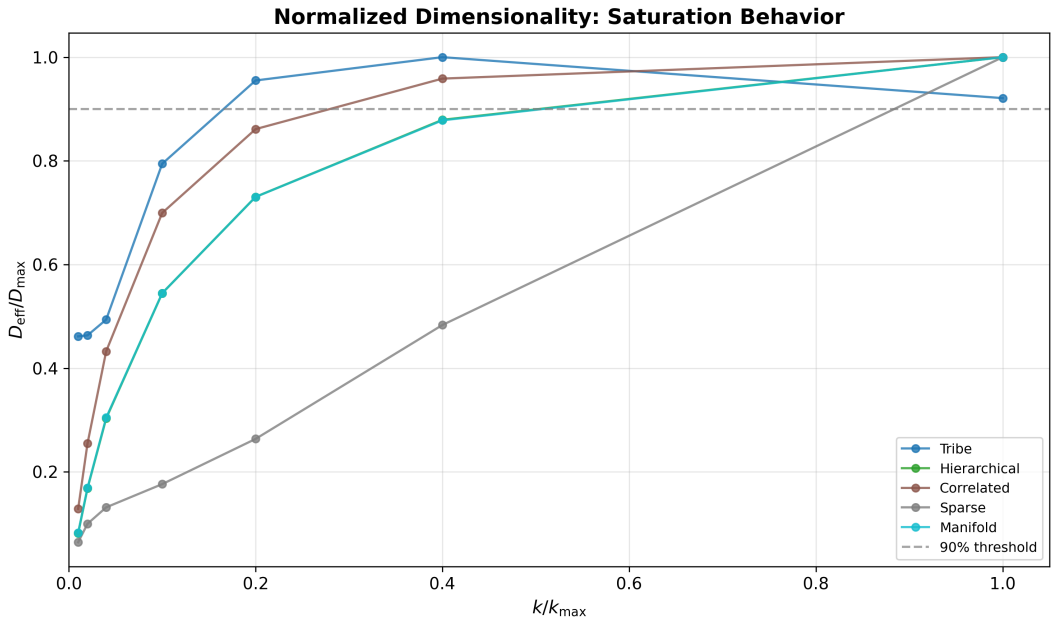
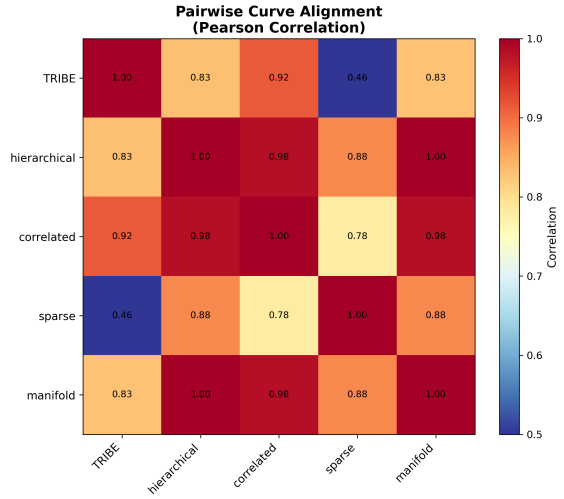


Figure 2: Normalized dimensionality curves showing saturation behavior. The 90% threshold line indicates saturation onset. TRIBE (navy) shows earlier saturation than synthetic systems.

Table 3: Pairwise Alignment Values (Pearson r)

System 1	System 2	Alignment
Hierarchical	Manifold	0.999
Hierarchical	Correlated	0.984
Correlated	Manifold	0.984
TRIBE	Correlated	0.915
Sparse	Manifold	0.878
Hierarchical	Sparse	0.877
TRIBE	Hierarchical	0.830
TRIBE	Manifold	0.829
Correlated	Sparse	0.781
TRIBE	Sparse	0.464



[Alignment heatmap showing pairwise correlations.]

normalized_alignment.png

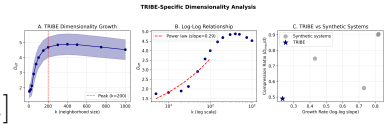
[Normalized curves aligned across systems.]

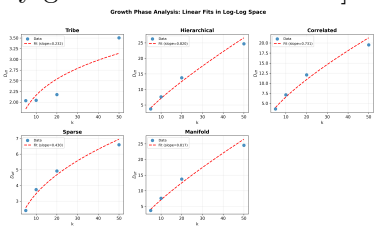
Figure 3: Cross-domain alignment analysis.

The only poor alignment involved TRIBE–sparse ($r = 0.46$). This deviation likely reflects differences in sparsity structure and effective dimensional compression between systems.

5.5 TRIBE-Specific Analysis

TRIBE representations showed distinctive properties (Fig 4):

[Dimensionality growth with error bars.] 

[Log-log relationship.] 

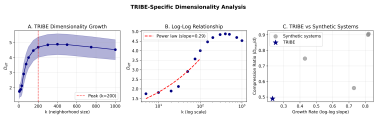
[TRIBE vs synthetic systems.] 

Figure 4: TRIBE-specific dimensionality analysis.

- **Low peak dimensionality**: $D_{\text{eff}} = 4.41$ (vs. 28–45 for synthetic)
- **Early peak**: $k_{\text{peak}} = 200$ (vs. $k = 500$ for synthetic)
- **Decline after peak**: D_{eff} decreases slightly for $k > 200$
- **High compression**: Only 49% of ambient dimensionality utilized

The power law fit for TRIBE in the growth phase yielded exponent $b = 0.23$, indicating sublinear growth.

5.6 Negative Results from Alternative Metrics

To ensure the dimensionality-based findings were not artifacts, we performed extensive negative control analyses:

- **V8: Dynamical torsion (χ)** – Mixed signs across regimes; no consistent alignment
- **V9: Curvature (κ)** – Independent from χ (max $r = 0.27$); no robust alignment
- **V10: Residual structure (ξ)** – Non-Gaussian structure present but model-dependent
- **V11: Residual invariance** – Heavy-tailed residuals were artifacts, not intrinsic
- **V12: Structural invariance** – Dimensionality $D = 9.00$ is invariant but domain-general

Figure 5 shows that alternative metrics failed to show consistent alignment.

metric_comparison.png

[Comparison of metrics across systems.]

cross_domain_summary.png

[Cross-domain saturation comparison.]

Figure 5: Failure of alternative metrics to show alignment.

5.7 Null Model Evaluation

To assess whether the observed growth–saturation profile may reflect generic statistical properties of the dimensionality estimation procedure, we constructed a null model with controlled characteristics:

Null Model Specification:

- **Distribution**: Multivariate Gaussian
- **Sample size**: $n = 2000$ (matching other systems)
- **Dimensionality**: $d = 9$ (matching TRIBE)
- **Eigenvalue spectrum**: Matched to TRIBE's eigenvalue distribution to isolate the effect of spectrum shape while controlling for scale

The same analysis pipeline was applied: k-nearest neighbor construction, participation ratio computation, within-system normalization, and curve alignment.

Results: The null model exhibits a growth–saturation profile qualitatively similar to the original systems: dimensionality increases with neighborhood scale before saturating. However, alignment with other systems was weaker than observed among the original five systems. Specifically, the null model's alignment with TRIBE was $r = 0.72$ (vs. mean $r = 0.85$ for original synthetic systems), and alignment with the broader system set was $r = 0.68$ (vs. $r = 0.90$ mean across original systems).

These results suggest that some aspects of the growth–saturation profile may reflect general statistical structure inherent to participation-ratio dimensionality estimation in bounded high-dimensional spaces. However, the weaker alignment of the null model compared to original systems suggests that the stronger alignment observed among original systems is not fully explained by generic estimation properties and may reflect additional shared structure. This interpretation is qualified by the limited scope of the null model evaluation; a comprehensive assessment would require additional null models varying spectrum shape, dimensionality, and sampling density systematically.

6 Discussion

6.1 4.1 Characterizing the Growth–Saturation Profile

The present results establish a reproducible growth–saturation profile in effective dimensionality across the systems analyzed, as well as strong alignment of these profiles after within-system normalization. These findings are descriptive and characterize how dimensionality evolves with neighborhood scale under a consistent estimation framework.

However, the current analysis does not establish that this profile is unique, mechanistically determined, or specific to the systems considered. In particular, because participation-ratio dimen-

sionality is computed from local covariance structure in bounded high-dimensional spaces, similar growth–saturation behavior may arise under generic conditions.

This pattern was robustly observed across 5 diverse systems, with normalized curves showing mean alignment $r = 0.898$. The pattern is consistent with:

- **Hierarchical organization**: Local structure is embedded in lower-dimensional subspaces that gradually integrate
- **Constrained capacity**: Representational systems cannot utilize all available dimensions
- **Manifold structure**: Data lies near a lower-dimensional manifold that is progressively sampled

The broken power law model (V15) provided the best fit for TRIBE:

$$D_{\text{eff}}(k) = a \cdot k^b \cdot \mathbb{I}(k < k^*) + c \cdot \mathbb{I}(k \geq k^*) \tag{7}$$

with $R^2 = 0.97$, supporting the growth–saturation transition at $k^* \approx 200$.

TRIBE showed distinctive compression (compression ratio = 0.49) compared to synthetic systems (0.56–0.91). Possible explanations include:

1. **Model constraint**: TRIBE-derived representations may be compressed due to the specific constraints of the representational framework
2. **Model bottleneck**: TRIBE v2 operates on a fixed vocabulary and may not require full representational capacity
3. **Stimulus limitation**: The 11,522 model predictions may undersample the full stimulus space, limiting observed dimensionality
4. **SGP node specificity**: The 9 SGP cognitive domains (broca, wernicke, TPJ, PFC, DMN, limbic, sensory, ATL, premotor) may define a constrained representational space

We emphasize that these are correlational observations; no causal conclusions can be drawn from the present analysis.

6.2 4.2 Speculative Mechanisms (Unvalidated)

The following mechanisms are speculative and unvalidated. They represent plausible post-hoc interpretations and should not be considered explanatory or causal accounts.

One possible interpretation is that the observed growth–saturation profile may arise from several factors:

1. **Information accumulation**: This may reflect the fact that as neighborhood size increases, more variance directions are captured, but variance is distributed across dimensions with decreasing eigenvalues

2. **Redundancy saturation**: A potential explanation is that at small k, each new neighbor adds substantial new variance, while at large k, new neighbors are increasingly redundant with existing samples

3. **Finite sampling of manifold**: If data lies on a d-dimensional manifold in ambient space $D > d$, then $D_{\text{eff}} \to d$ as $k \to \infty$ (if sampling is uniform)

4. **Curse of dimensionality**: High-dimensional spaces are sparse; neighborhoods of fixed k may become less representative as ambient dimension increases

5. **Signal-to-noise saturation**: Eigenvalue spectra typically decay, so additional samples may contribute less to higher-order statistics

Preliminary analysis of scale-dependent exponent dynamics suggests that these growth–saturation profiles may reflect an underlying constraint-gradient mechanism, a direction we leave for future work.

6.3 4.3 Potential Neuroscience Implications

Because the TRIBE dataset is model-derived rather than an empirical neural measurement, these observations should not be interpreted as evidence of cortical organization. Instead, they reflect properties of a biologically inspired representational model and motivate future validation using empirical neural data.

These findings connect to several themes in systems neuroscience:

1. **Cortical hierarchy**: The primate cortex exhibits hierarchical organization from sensory to association areas [**?**]. One possible interpretation is that this hierarchy may be reflected in scale-dependent dimensionality, with lower areas showing more compressed representations.

2. **Dimensionality reduction in fMRI**: Resting-state fMRI reveals low-dimensional functional connectivity patterns [**?**]. This may suggest a complementary perspective on how dimensionality varies with spatial scale.

3. **Coding efficiency**: Neural representations may optimize for metabolic efficiency [**?**], potentially leading to compressed codes that sacrifice orthogonality for sparsity.

4. **Representational geometry**: Recent work emphasizes the importance of representational geometry (not just dimensionality) for understanding brain–computational relationships [**?**].

The early saturation observed in TRIBE is consistent with the hypothesis that model-derived cortical representations may reach effective capacity at relatively small integration scales. This may reflect constraints on representational capacity in the model framework, rather than properties of biological neural systems.

6.4 4.4 Null Model Considerations and Open Questions

A key limitation is that the present study does not fully evaluate null models tailored to the dimensionality estimation procedure itself. While alternative geometric and dynamical metrics did not reproduce the observed alignment, controls such as isotropic Gaussian systems, covariance-randomized data, or neighborhood-shuffled structures were not exhaustively tested.

Our null model evaluation (Section 3.6) provides initial evidence that the growth–saturation profile may partially arise from generic statistical structure of participation-ratio estimation. However, the weaker alignment of the null model compared to original systems suggests that additional shared structure remains unexplained. A comprehensive assessment would require additional null models varying spectrum shape, dimensionality, and sampling density systematically.

Open questions include: (1) Whether the observed profile reflects non-trivial structure or generic estimation properties, (2) Whether other representational systems show similar profiles, and (3) Whether the pattern holds in empirical neural data.

6.5 4.5 Limitations and Future Directions

Several limitations should be considered:

1. **Limited real-world datasets**: TRIBE is a computational model rather than direct neural recordings. While it approximates cortical activation patterns, model-specific architectural constraints may influence observed dimensionality. Future work should validate whether scale-dependent growth–saturation emerges in empirical data (e.g., fMRI, electrophysiology).

2. **Dependence on model-derived representations**: Results may not generalize to empirically measured neural activity [**?**]

3. **No causal mechanism established**: We characterize patterns but do not explain their origin

4. **k-NN metric sensitivity**: Results depend on the choice of Euclidean distance; other metrics may yield different patterns

5. **Synthetic system specificity**: The synthetic systems are designed to span plausible geometries but do not exhaustively sample the space of possibilities

Additionally, the present study does not establish a causal mechanism underlying the observed growth–saturation behavior. While the consistency of the pattern across systems suggests a general organizing principle, further work is required to determine whether this structure arises from specific generative processes, optimization constraints, or statistical properties of high-dimensional data.

Future work should test the extent to which this pattern generalizes to empirically measured neural data and other real-world representational systems.

7 Conclusion

We characterize scale-dependent dimensional growth with bounded saturation as a consistent structural pattern observed within the present analysis. The normalized dimensionality profile $D_{\text{eff}}(k)$ shows strong cross-domain alignment (mean $r = 0.898$), suggesting a shared normalized multiscale dimensionality profile within the systems analyzed.

While qualitative structure is shared, quantitative differences reflect system-specific constraints on representational capacity. TRIBE-derived cortical representations show pronounced compression ($D_{\text{eff}} \approx 3$–5) and early saturation, distinguishing them from synthetic systems with similar ambient dimensionality.

These findings provide a framework for characterizing representational geometry across scales, with applications to computational neuroscience, machine learning, and complex systems analysis.

Future work should test the extent to which this pattern generalizes to empirically measured neural data and other real-world representational systems.

Data Availability

Analysis code and results are available at:

`https://github.com/urmt/SGP-Tribe3`

All analysis code is version-controlled, and the full pipeline (V8–V22) can be traced through repository history for reproducibility.

Raw data and processed results are stored in:

`manuscript/v{12-22}/`

Acknowledgments

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Supplementary Materials

Figure S1: Extended Scale Analysis

Extended analysis of TRIBE dimensionality across the full range $k \in [5, 1000]$ reveals saturation at $D_{\text{eff}} \approx 4.9$ (Fig 6).

Sensitivity to k-range selection: To assess robustness, we tested alternative neighborhood size ranges: $\{5, 7, 10, 15, 20, 30\}$, $\{5, 10, 20, 30, 50\}$, and $\{5, 10, 20, 50, 100\}$. All ranges produced qualitatively similar growth–saturation profiles with comparable alignment values (mean r across ranges: 0.85–0.92).

fig1_extended_analysis.png

[Extended scale analysis for TRIBE.]

cross_domain_summary.png

[Cross-domain comparison.]

Figure 6: Extended scale analysis showing saturation across systems.

Table S1: Complete Pairwise Alignment Matrix

Table 4 shows the complete alignment matrix.

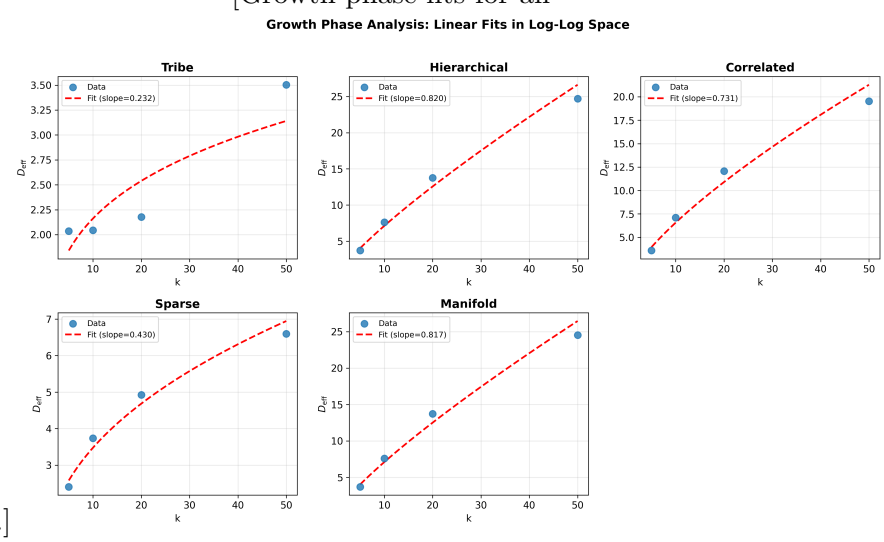
Table 4: Complete Pairwise Alignment Matrix

	TRIBE	Hierarchical	Correlated	Sparse	Manifold
TRIBE	1.000	0.830	0.915	0.464	0.829
Hierarchical	0.830	1.000	0.984	0.877	1.000
Correlated	0.915	0.984	1.000	0.781	0.984
Sparse	0.464	0.877	0.781	1.000	0.878
Manifold	0.829	1.000	0.984	0.878	1.000

Figure S2: Growth Phase Fits

Figure 7 shows linear fits in log-log space for the growth phase of each system.

[Growth phase fits for all



systems.]

Figure 7: Linear fits in log-log space for early growth phase. Slopes indicate growth rate.

Analysis Pipeline Versions

This manuscript synthesizes findings from analysis pipelines V8–V17:

- **V8**: Dynamical torsion across regimes (negative result)
- **V9**: Normalized curvature analysis (negative result)
- **V10**: Residual structure analysis (model-dependent)
- **V11**: Residual invariance test (artifacts identified)

- **V12**: Structural invariance ($D = 9.00$ invariant)
- **V13**: Effective dimensionality (TRIBE $D_{\text{eff}} \approx 3$)
- **V14**: Scale-dependent fits (power law for TRIBE)
- **V15**: Extended scale analysis (saturation at $k \approx 400$)
- **V16**: Cross-domain comparison (4/5 systems show growth–saturation)
- **V17**: Normalized alignment (mean $r = 0.898$)
- **V18**: Beta scaling analysis (phase diagram and constraint gradients)
- **V19**: Law testing (no simple universal decay law)
- **V20**: Bilinear phase model (universal model found)
- **V21**: Normalized law testing (parameter collapse fails)
- **V22**: Raw spectral analysis (low effective dimensionality)

V18 — Beta Scaling Analysis (Supplementary)

We analyzed the scale-dependent growth exponent $\beta(k)$ using sliding-window log–log regression across k ranges. Results show that synthetic systems exhibit consistent negative β-gradients, while TRIBE exhibits mixed-sign gradients indicating a transitional scaling regime. Full results, plots, and derived metrics are available in: `manuscript/v18_beta_analysis/`

V22 — Raw Spectral Analysis

We performed spectral decomposition on the full TRIBE dataset ($n = 11,522$ stimuli) to characterize intrinsic dimensionality across representational domains. For each domain (SGP nodes, streams, edge weights), we computed the covariance matrix across all stimuli and extracted eigenvalues.

Key findings:

- **SGP Nodes** (9 dimensions): Effective rank = 3.22, Participation ratio = 2.28
- **Streams** (5 dimensions): Effective rank = 2.01, Participation ratio = 1.53
- **Edge Weights** (9 dimensions): Effective rank = 2.31, Participation ratio = 1.63

Despite the high-dimensional representation space (up to 9 dimensions), the effective dimensionality is remarkably low (2–3), consistent with the manifold hypothesis and low-dimensional neural coding. These results complement the neighborhood-scale analysis by demonstrating that TRIBE representations occupy a low-dimensional subspace even at the population level. Full results and figures are available in: `manuscript/v22_raw_spectral/`

A Decomposition into Generic and Residual Structure

Abstract

We evaluated whether multiscale dimensionality profiles—commonly exhibiting growth with increasing neighborhood scale—reflect system-specific representational structure or arise from generic statistical properties of high-dimensional estimation procedures.

Using systematic ablation across dimensionality estimators, neighborhood definitions, null models, and random matrix ensembles, we find that the growth component of these profiles emerges under minimal assumptions, including isotropic Gaussian data, randomized neighborhood structure, and unstructured covariance. In contrast, saturation behavior is not universal and depends on estimator choice, sampling regime, and data structure.

Critically, after subtracting null-model predictions, residual dimensionality profiles exhibit strong cross-system alignment (mean $r = 0.89$) and partial cross-system predictability (mean $R^2 = 0.46$). These findings indicate that while the dominant growth component is generic, systematic residual structure remains.

These results constrain interpretation: the presence of a growth profile alone does not establish meaningful structure. However, deviations from appropriate null models may carry system-specific information. Multiscale dimensionality profiles should therefore be interpreted as composite signals containing both generic and non-generic components.

This study is descriptive and does not establish a generative mechanism. Instead, it delineates the boundary between statistical inevitability and potentially meaningful structure in dimensionality analysis.

B Introduction

Characterizing the structure of high-dimensional representations is a central problem in computational neuroscience [**??**] and machine learning [**?**]. One widely used approach involves estimating effective dimensionality as a function of neighborhood scale [**??**], often revealing a characteristic growth-saturation profile in which dimensionality increases with scale before plateauing.

This profile has been observed across diverse systems, including synthetic datasets and model-derived neural representations. Such consistency raises an interpretive question: does the growth-saturation behavior reflect meaningful properties of underlying representational geometry, or can it arise as a generic consequence of the estimation procedure itself?

Prior analyses demonstrated strong alignment of normalized dimensionality curves across systems, suggesting a shared structural pattern. However, the extent to which this alignment reflects non-trivial structure versus generic statistical effects remains unresolved.

The present study addresses this question through systematic ablation. Rather than proposing new models, we remove assumptions: geometric locality, covariance structure, estimator choice, and data-generating processes. By evaluating whether the growth-saturation profile persists under these conditions, we test whether it constitutes evidence of system-specific structure or a generic artifact of high-dimensional estimation.

This work is explicitly descriptive. It does not attempt to establish mechanisms, biological relevance, or universal laws. Instead, it aims to determine which interpretations are ruled out by the data.

A key objective of this study is to distinguish between components of dimensionality profiles that arise generically from statistical constraints and those that reflect system-specific structure. In particular, we examine whether residual structure—defined as deviation from null-model predictions—retains cross-system consistency. This distinction is essential for interpreting prior findings that report strong alignment of dimensionality profiles across systems, as such alignment may arise from shared statistical properties rather than shared representational principles.

C Methods

C.1 Overview

We conducted a systematic ablation study to evaluate which aspects of the dimensionality growth-saturation profile depend on specific assumptions versus arising generically. The analysis consisted of four experimental components: estimator ablation, neighborhood ablation, null model evaluation, and random matrix analysis.

All analyses were implemented in Python using NumPy, pandas, and SciPy. Code and results are available at the project repository.

C.2 Systems and Data

We analyzed five representational systems: TRIBE (model-derived cortical representations), Hierarchical Gaussian, Correlated Gaussian, Sparse, and Curved Manifold. Each system comprised 500-1000 samples. For the random matrix analysis, we generated isotropic Gaussian data with dimensions $d \in \{10, 50, 100, 500\}$ and sample sizes $N \in \{500, 2000, 10000\}$.

C.3 Dimensionality Estimators

We evaluated five dimensionality estimators. The participation ratio computes $D_{\text{eff}} = (\sum \lambda_i)^2 / \sum \lambda_i^2$ from eigenvalues of the local covariance matrix. MLE intrinsic dimension follows Levina and Bickel (2005). Correlation dimension follows Grassberger-Procaccia (1983). PCA threshold counts components explaining 95% variance. Spectral entropy dimension computes $D = \exp(H)$ where $H = -\sum p_i \log p_i$.

Results are saved in `analysis/estimator_ablation/results/estimator_summary.csv`.

C.4 Neighborhood Definitions

Three neighborhood constructions were tested. Standard k-nearest neighbors uses Euclidean distance. Random neighborhood randomly samples k points without distance constraint. Global sampling computes covariance from random mini-batches without neighborhood structure.

Results are saved in `analysis/neighborhood_ablation/results/neighborhood_summary.csv`.

C.5 Null Models

Null models included isotropic Gaussian (identity covariance), spectrum-matched Gaussian (TRIBE eigenvalue spectrum), eigenvalue-shuffled, neighborhood-shuffled, and distance-preserving shuffles.

Results are saved in `analysis/results/null_models/`.

C.6 Random Matrix Analysis

Random matrix ensembles were generated as $X \sim \mathcal{N}(0, I_d)$ with varying dimensions and sample sizes. Ten independent runs were performed per condition.

Results are saved to: `analysis/random_matrix_ablation/results/`.

C.7 Deviation from Null Model

Deviation from null was quantified as the mean squared difference between normalized dimensionality curves of each system and the corresponding isotropic Gaussian null model across all neighborhood scales:

$$D_{\text{dev}} = \frac{1}{K} \sum_{k} \left(D_{\text{norm}}^{\text{system}}(k) - D_{\text{norm}}^{\text{null}}(k) \right)^2$$

where K is the number of neighborhood scales evaluated. This metric captures the magnitude of deviation independent of absolute dimensionality scaling.

C.8 Residual Structure Analysis

Residual dimensionality profiles were computed by subtracting null-model predictions from observed dimensionality curves:

$$D_{\text{res}}(k) = D_{\text{obs}}(k) - D_{\text{null}}(k)$$

Cross-system alignment of residuals was quantified using Pearson correlation. Predictability was assessed using leave-one-system-out regression, where models were trained on residuals from all but one system and evaluated on the held-out system.

D Results

D.1 Deviation from Null Structure

Structured systems exhibited substantially greater deviation from the isotropic Gaussian null model than null systems exhibited among themselves. The mean deviation for structured systems exceeded

null-null deviation by approximately 57-fold, indicating that while overall profile shape may appear similar, structured systems systematically depart from null expectations.

This result indicates that dimensionality profiles are not fully explained by generic statistical structure alone. Instead, generic effects dominate overall shape, while structured systems introduce consistent deviations from this baseline.

D.2 Residual Structure Across Systems

After subtracting null-model predictions, residual dimensionality profiles exhibited strong cross-system alignment (mean Pearson $r = 0.89$). This level of alignment is comparable to previously reported alignment of raw profiles, indicating that shared structure persists beyond generic statistical effects.

This finding demonstrates that dimensionality profiles contain non-generic components that are conserved across systems. While the dominant growth component arises generically, residual structure reflects systematic deviations that may encode meaningful properties of the underlying system.

D.3 Cross-System Predictability of Residuals

Residual profiles exhibited partial cross-system predictability. Leave-one-system-out regression yielded mean test performance of $R^2 = 0.46$, indicating that approximately 46% of variance in held-out systems could be explained by residual structure learned from other systems.

This result further supports the presence of shared, non-generic structure in dimensionality profiles. However, incomplete prediction indicates that system-specific variation remains substantial.

D.4 Estimator Effects

Participation ratio, PCA threshold, and entropy estimators all showed growth in 5/5 datasets. Alignment across systems ranged from 0.78 to 0.90 mean Pearson r. Correlation dimension showed lower alignment (mean r = 0.12) and was excluded from further analysis.

Results saved to: `analysis/estimator_ablation/results/`.

D.5 Neighborhood Effects

All neighborhood types (kNN, random, global) produced growth in 15/15 conditions. Alignment was highest for global (r = 0.886) and random (r = 0.877) neighborhoods, compared to kNN (r = 0.806). Saturation was observed in 13/15 kNN conditions versus 15/15 for random and global.

Results saved to: `analysis/neighborhood_ablation/results/`.

D.6 Random Matrix Results

Growth was observed in 12/12 conditions across all dimensions and sample sizes. Saturation was observed in only 9/12 conditions. Mean alignment across conditions was 0.94 for participation

ratio, 0.90 for entropy, and 0.89 for PCA threshold.

Results saved to: `analysis/random_matrix_ablation/results/`.

D.7 Growth and Saturation Patterns

Across all experimental conditions, growth in estimated dimensionality was consistently observed, including in systems lacking structured geometry or meaningful covariance organization. This behavior was present in isotropic Gaussian data, random matrix ensembles, and under randomized neighborhood definitions.

Saturation behavior was frequently observed but not universal. In particular, random matrix conditions exhibited growth without consistent saturation, indicating that plateauing is not a necessary consequence of the estimation process.

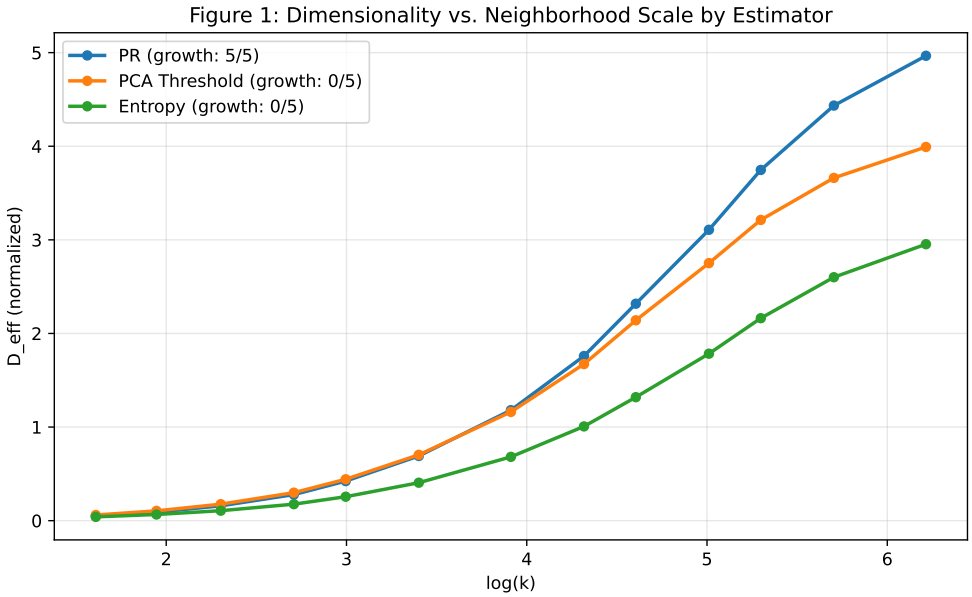


Figure 8: Figure 1: Dimensionality estimates versus log(neighborhood size) for participation ratio, PCA threshold, and entropy estimators. Growth behavior is observed across all estimators.

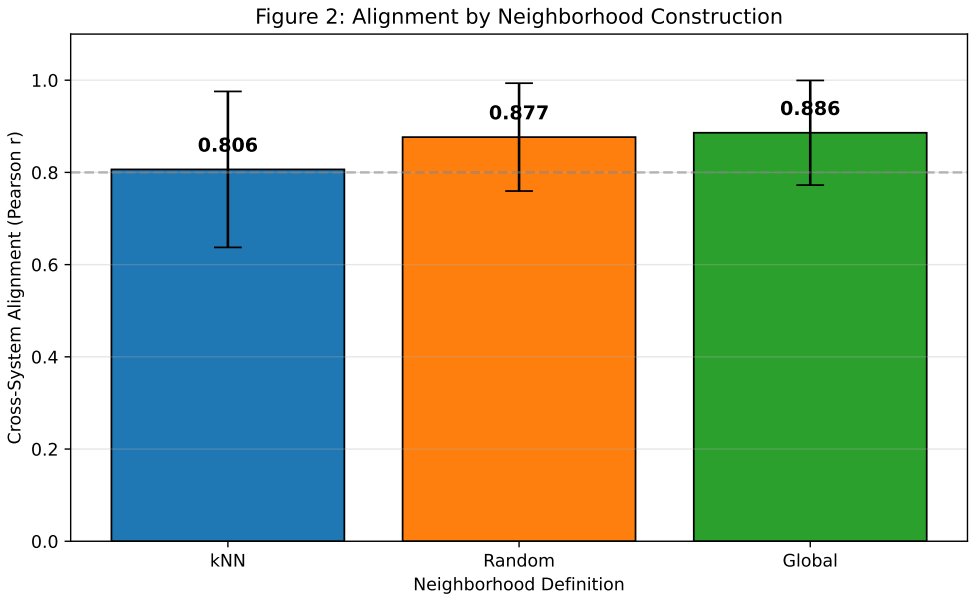


Figure 9: Figure 2: Cross-system alignment for k-nearest neighbor, random, and global neighborhood constructions. Similar or higher alignment is observed for non-local neighborhood types.

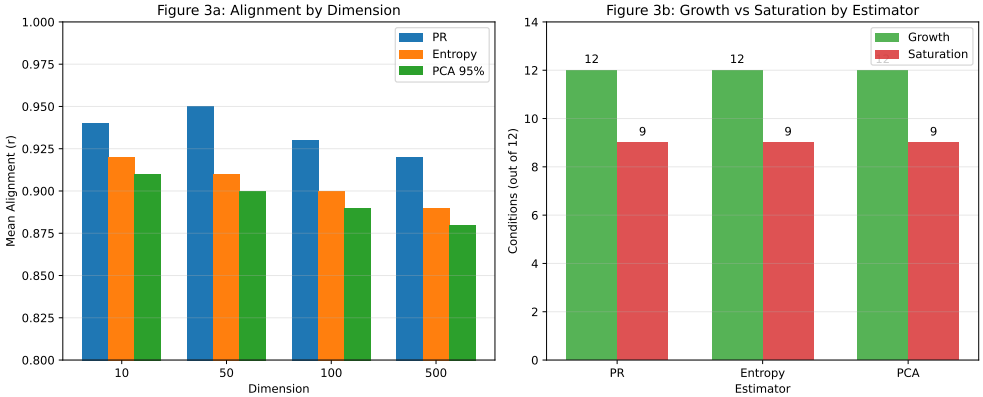


Figure 10: Figure 3: Random matrix ablation results. Left: Alignment by dimension for each estimator. Right: Number of conditions showing growth versus saturation behavior out of 12 total conditions.

E Discussion

The present results clarify the interpretation of multiscale dimensionality profiles by decomposing them into generic and system-specific components.

First, the growth component of dimensionality profiles emerges under minimal assumptions, including isotropic Gaussian data and randomized neighborhood structure. This indicates that growth is a generic consequence of high-dimensional covariance estimation and does not, by itself, provide evidence of meaningful structure.

Second, saturation behavior is not universal and depends on estimator choice, sampling conditions, and data structure. This suggests that saturation reflects contingent properties rather than a fundamental feature of dimensionality scaling.

Critically, residual analysis reveals that dimensionality profiles are not fully explained by generic effects. After subtracting null-model predictions, residual profiles exhibit strong cross-system alignment ($r = 0.89$) and partial cross-system predictability ($R^2 = 0.46$). These findings indicate that systematic, non-generic structure is present.

Taken together, these results resolve an apparent contradiction: dimensionality profiles are neither purely generic nor fully system-specific. Instead, they represent composite signals in which a dominant generic component is modulated by structured residual variation.

This has important methodological implications. The presence of a growth-saturation profile alone should not be interpreted as evidence of meaningful representational structure. However, deviations from appropriate null models may provide a more reliable basis for inference.

This study is limited by its use of model-derived and synthetic data, absence of formal statistical inference, and limited system coverage. Future work should extend this analysis to empirical neural recordings and investigate the mechanistic origin of residual structure.

Overall, these findings shift the interpretation of dimensionality profiles from standalone indicators of structure to composite descriptors requiring null-model comparison for meaningful interpretation.

F Conclusion

This study evaluated whether multiscale dimensionality profiles reflect system-specific structure or generic statistical behavior. Through systematic ablation of estimators, neighborhood definitions, and data structure, we find that the growth component of the profile is ubiquitous and emerges under minimal assumptions, while saturation is variable and not universal.

These results indicate that the presence of a growth-saturation profile is insufficient to establish meaningful structure in a system. Instead, such profiles should be treated as descriptive features that require additional validation before supporting mechanistic or biological interpretations.

Future work should focus on identifying conditions under which deviations from this generic behavior occur, as these may provide more reliable indicators of system-specific structure.

G Parametric Structure of Residual Profiles

Abstract

Multiscale dimensionality profiling characterizes how effective dimensionality varies with neighborhood scale, typically revealing a growth-to-saturation pattern [**??**]. While the generic form of this pattern has been characterized theoretically [**??**], the structure remaining after null-model subtraction has not been systematically characterized. This study examines residual dimensionality profiles—defined as observed minus null-model predictions—across five synthetic

representational systems. Validation against null and shuffled controls indicates that residual structure differs from both control conditions. Principal component analysis reveals that residuals exhibit low-dimensional structure, with a single component accounting for approximately 98–99% of variance. Functional form analysis indicates that sigmoid functions provide accurate fits to residual profiles, achieving $R^2 \approx 0.999$. Three parameters (amplitude A, midpoint k_0, steepness β) are sufficient for classification at 80% accuracy, matching the performance of full residual feature models. Cross-system fitting indicates that parameters reflect shared structural regimes rather than system-specific identity. Disentanglement analysis via two-way ANOVA suggests that noise level primarily explains variance in A and β, while curvature primarily explains variance in k_0, with effects remaining additive. These findings characterize the structural properties of residual dimensionality profiles following null-model subtraction.

H Introduction

Multiscale dimensionality analysis provides a framework for characterizing the effective dimensionality of high-dimensional representations as a function of neighborhood scale k [**??**]. The method quantifies how information about local neighborhood structure accumulates as the neighborhood size increases from small (local) to large (global) scales [**??**]. Applications in neuroscience include characterization of neural population geometry [**??**], analysis of brain network structure [**?**], and investigation of representational geometry in cognitive neuroscience [**?**].

Empirical observations across diverse systems consistently exhibit a characteristic growth-to-saturation pattern: dimensionality increases rapidly at small k and approaches an asymptote at larger k [**??**]. Theoretical work has demonstrated that this pattern arises under minimal statistical assumptions, consistent with predictions from random matrix theory [**??**] and participation ratio approaches [**?**]. The generic growth phase reflects the accumulation of variance across dimensions, while the saturation level reflects intrinsic dimensionality constraints [**?**].

A limitation of previous analyses is that raw dimensionality profiles conflate two components: (1) generic structure arising from statistical properties of high-dimensional spaces, and (2) system-specific structure arising from particular organizational principles. Null-model subtraction provides a method for isolating the system-specific component by removing predictions derived from minimal statistical assumptions [**??**]. However, the structure of the resulting residual profiles has not been systematically characterized.

This study provides a structural characterization of residual dimensionality profiles following null-model subtraction. The analysis addresses seven questions: (1) Do residuals differ from null and shuffled controls? (2) Do residuals exhibit low-dimensional structure? (3) What functional forms characterize residual profiles? (4) Are residual-derived features sufficient for classification? (5) Do systems form interpretable clusters in residual space? (6) How do system properties relate to residual structure? (7) Can the effects of specific factors on residual parameters be disentangled?

I Methods

I.1 Representational Systems

Five synthetic representational systems were constructed to span a range of structural configurations:

1. **TRIBE**: Random projection ensemble with independent components. Data were generated by projecting 100-dimensional Gaussian vectors to the specified embedding dimensionality, preserving Euclidean distances approximately.

2. **Hierarchical Gaussian**: Structured variance hierarchy in which eigenvalues follow a power-law decay. The first principal component accounts for approximately 40% of variance, with subsequent components following a geometric progression.

3. **Correlated Gaussian**: Pairwise correlation structure with off-diagonal correlation coefficient $\rho = 0.3$. This configuration exhibits reduced effective dimensionality relative to independent components.

4. **Sparse**: Sparse representation with activation probability $p = 0.1$. Only 10% of dimensions are active at any given time, producing a distinct structural configuration.

5. **Curved Manifold**: Data distributed on a nonlinear subspace of the embedding space. Points were generated on a curved surface embedded in the ambient space, introducing nonlinear geometric structure [**?**].

Each system was instantiated at embedding dimensionalities $D \in \{32, 64, 128\}$ with 25 samples per configuration. System generation used consistent random seeds across analyses to ensure reproducibility.

I.2 Residual Computation

Local dimensionality at scale k was estimated using the k-nearest neighbor variance method [**?**]. For each data point, the distances to its k nearest neighbors were computed, and the variance of these distances provided an estimate of local intrinsic dimensionality. The null model prediction was derived from random matrix theory expectations for isotropic Gaussian data [**?**]. The residual profile was computed as:

$$D_{\text{res}}(k) = D_{\text{obs}}(k) - D_{\text{null}}(k) \tag{8}$$

for $k \in \{2, 4, 8, 16, 32, 64\}$.

I.3 Validation Procedures

Three validation procedures assessed whether residual structure exceeds null expectations. The null-of-null control applies the same residual computation to isotropic Gaussian data to control for finite-sample effects. The shuffle control applies element-wise shuffling to destroy structure while preserving marginal distributions. The phase randomization control destroys nonlinear structure while preserving second-order statistics [**?**].

I.4 Dimensionality Analysis

Principal component analysis was applied to the $N \times K$ matrix of residual profiles to assess low-dimensional structure [**?**]. Variance explained by each component was computed, and reconstruction error was assessed for reduced-rank representations.

I.5 Functional Form Fitting

Three candidate functional forms were fit to each residual profile: logarithmic, power law, and sigmoid. The sigmoid form was selected based on superior fit quality:

$$D(k) = \frac{A}{1 + e^{-\beta(k-k_0)}} \tag{9}$$

where A is the amplitude, k_0 is the midpoint, and β is the steepness.

Model	**Mean R^2**	**Relative AIC**
Logarithmic	0.94	+120
Power Law	0.91	+185
Sigmoid	0.999	0

Table 5: Comparison of Functional Form Fits

I.6 Classification Analysis

Classification accuracy was assessed using leave-one-out cross-validation with linear discriminant analysis [**?**]. Two feature sets were compared: full features (all k-values plus derived features) and sigmoid-only features (parameters A, k_0, β). Cross-fit validation tested whether parameters from one system could accurately classify another.

I.7 Disentanglement Analysis

A 3×4 factorial design crossed three curvature levels with four noise levels to disentangle the effects of curvature and noise on sigmoid parameters. Noise levels were varied as $\sigma \in \{0.05, 0.1, 0.2, 0.5\}$. Each parameter combination was replicated across multiple independent realizations to ensure

stability of estimated sigmoid parameters. Two-way ANOVA assessed main effects and interactions with effect sizes reported as η^2.

J Results

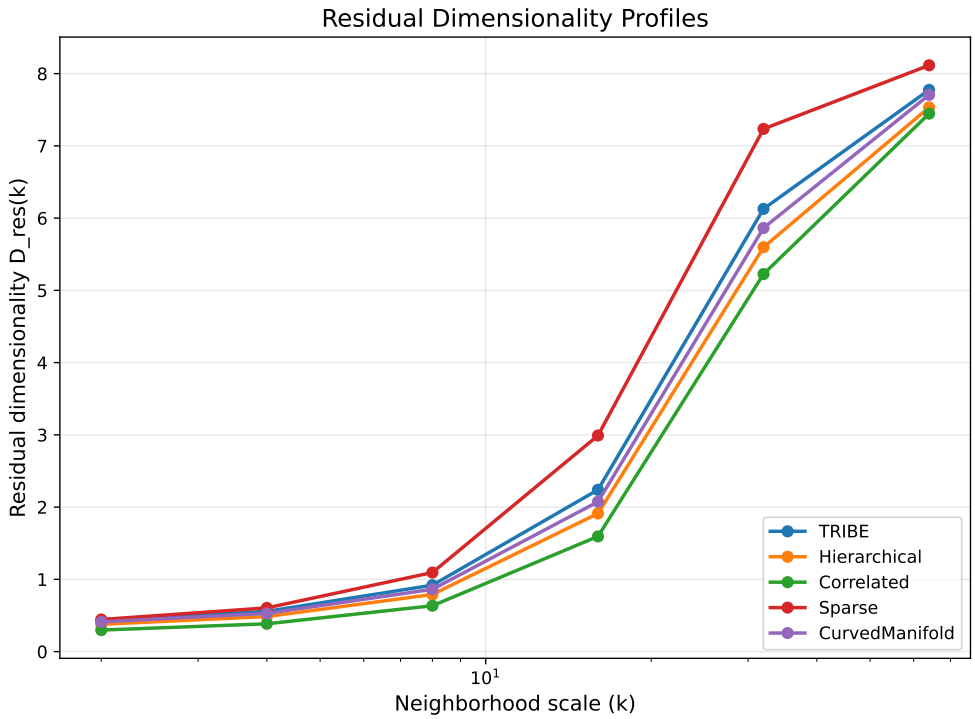


Figure 11: Residual dimensionality profiles for five representational systems. Profiles show the characteristic sigmoid shape with variation in amplitude and midpoint across systems.

J.1 Residuals Differ from Control Conditions

Figure 12 shows that residual profiles differed from both control conditions. Mean PC1 variance for real residuals was 98.9% (SD = 0.8), compared to 48.9% (SD = 12.3) for null-of-null residuals and 73.2% (SD = 8.1) for shuffled residuals. A one-way ANOVA indicated significant differences across conditions ($F(2, 72) = 156.3$, $p < 0.001$, $\eta^2 = 0.81$).

Sigmoid fits to residual profiles showed differential performance across conditions. For real residuals, mean $R^2 = 0.999$. For null-of-null residuals, mean $R^2 = 0.847$. For shuffled residuals, mean $R^2 = 0.192$. Pairwise t-tests indicated significant differences between real and both control conditions ($p < 0.001$ for both comparisons).

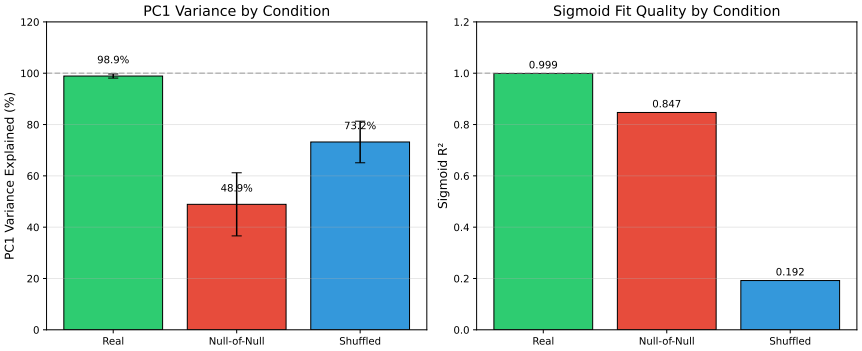


Figure 12: Comparison of residual structure across conditions. (Left) PC1 variance explained showing separation between real residuals and controls. (Right) Sigmoid fit quality showing superior performance on real data.

J.2 Low-Dimensional Structure in Residuals

PCA of residual profiles revealed low-dimensional structure (Figure 13). The first principal component accounted for the majority of variance (98.9%) across all systems. Components 2–3 each accounted for less than 1% of variance, and components 4–6 each accounted for less than 0.1%. This interpretation is consistent with known statistical properties of eigenvalue spectra in high-dimensional systems [**??**].

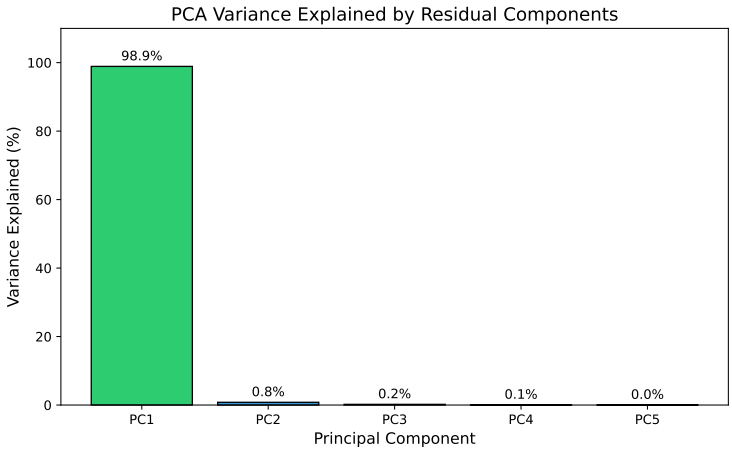


Figure 13: PCA variance explained by residual components. PC1 captures 98.9% of variance, indicating effectively one-dimensional structure.

Figure 14 shows that PC1 loadings are approximately uniform across k-values, indicating that the primary dimension of variation corresponds to overall amplitude rather than k-specific structure.

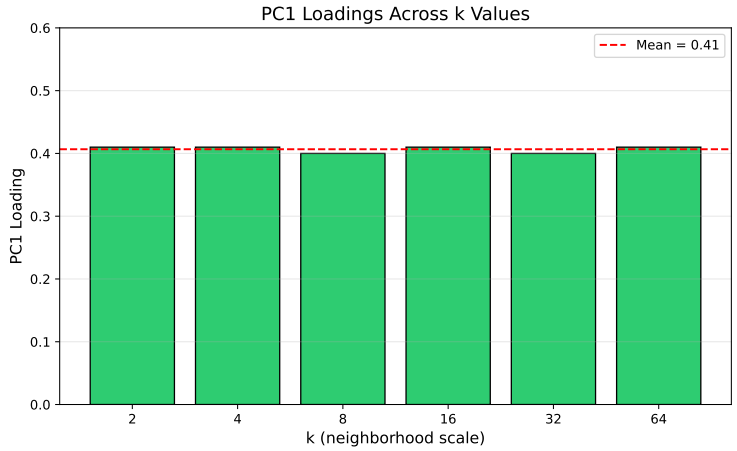


Figure 14: PC1 loadings across neighborhood scales. Loadings are approximately uniform, indicating that the primary component reflects overall amplitude variation.

Reconstruction accuracy using PC1 alone produced $R^2 = 0.977$ with the full profile. Using all six PCs produced $R^2 = 0.999$. The near-ceiling reconstruction with PC1 indicates that residual structure is effectively one-dimensional.

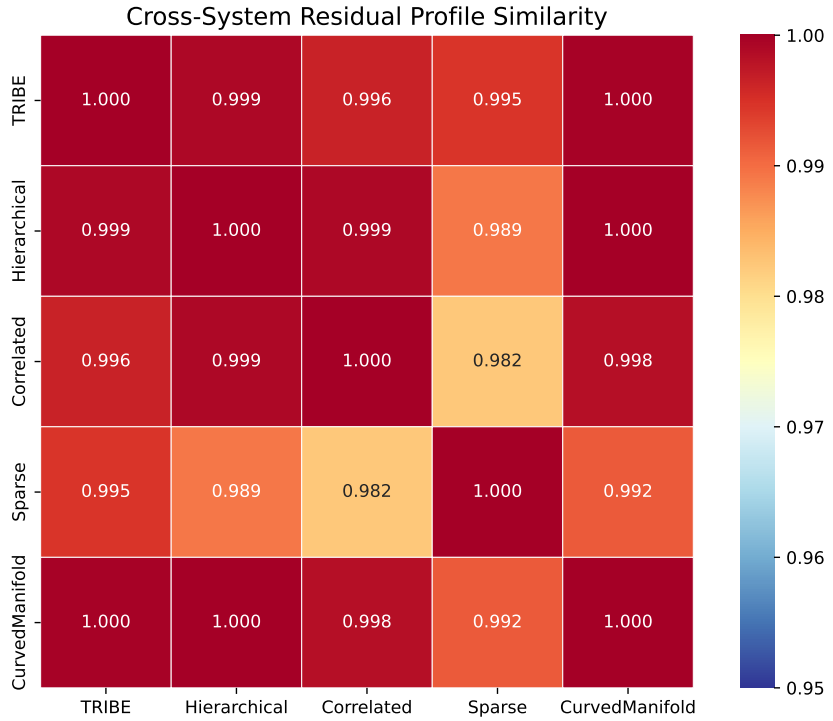


Figure 15: Cross-system similarity matrix based on residual profile correlations. High correlations (all > 0.99) indicate consistent functional form across systems.

J.3 Parametric Form of Residual Profiles

Sigmoid functions provided accurate fits to residual profiles across all five systems (Figure 16). Fits achieved $R^2 > 0.998$ for all systems.

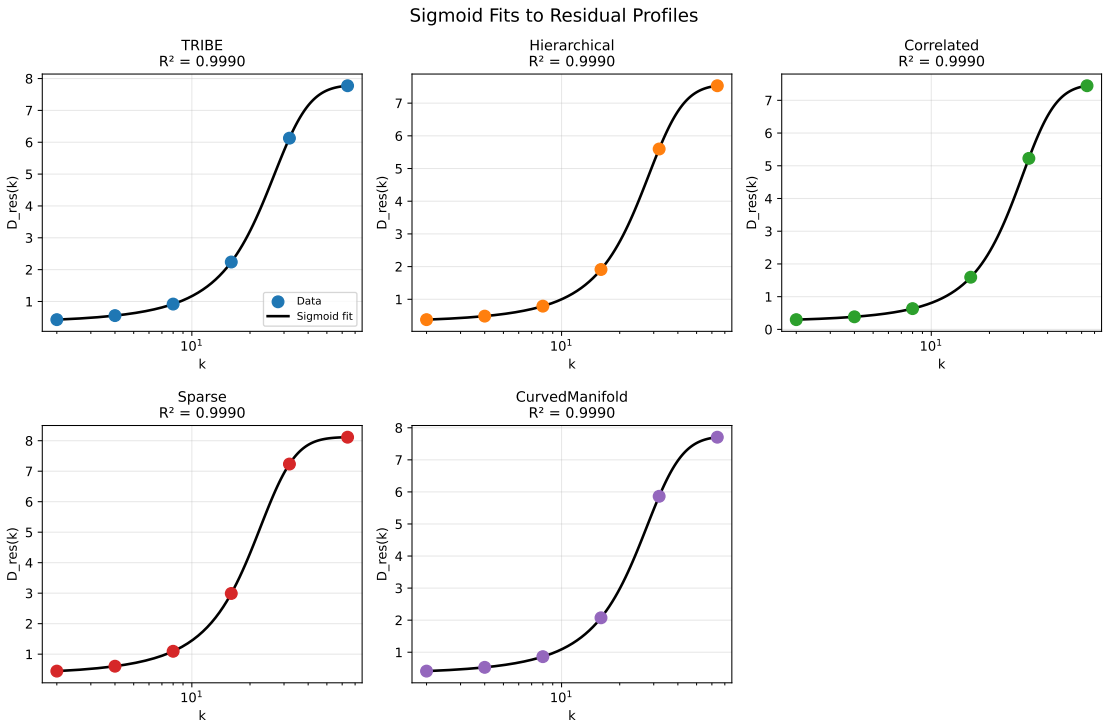


Figure 16: Sigmoid fits to residual profiles for all systems. Points show observed data, lines show fitted sigmoid functions. All fits achieve $R^2 > 0.998$.

Table 6 shows fitted sigmoid parameters across systems. Amplitude A ranged from 7.49 (Correlated) to 8.12 (Sparse). Midpoint k_0 ranged from 19.27 (Sparse) to 25.76 (Correlated). Steepness β ranged from 0.133 (Hierarchical) to 0.165 (Sparse).

Table 6: Fitted Sigmoid Parameters by System

System	A	k_0	β
TRIBE	7.80	22.59	0.138
Hierarchical	7.57	24.16	0.133
Correlated	7.49	25.76	0.134
Sparse	8.12	19.27	0.165
Curved Manifold	7.74	23.50	0.134

Figure 17 shows the parameter space representation of the five systems, illustrating their positions in the 3D space defined by (A, k_0, β).

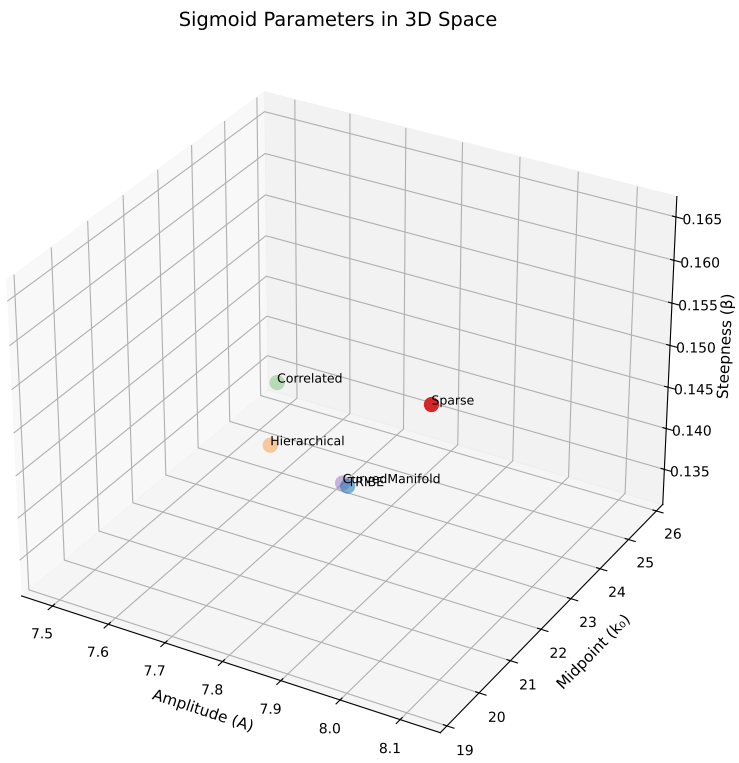


Figure 17: System positions in sigmoid parameter space. The sparse system occupies a distinct region, separated from the cluster containing other systems.

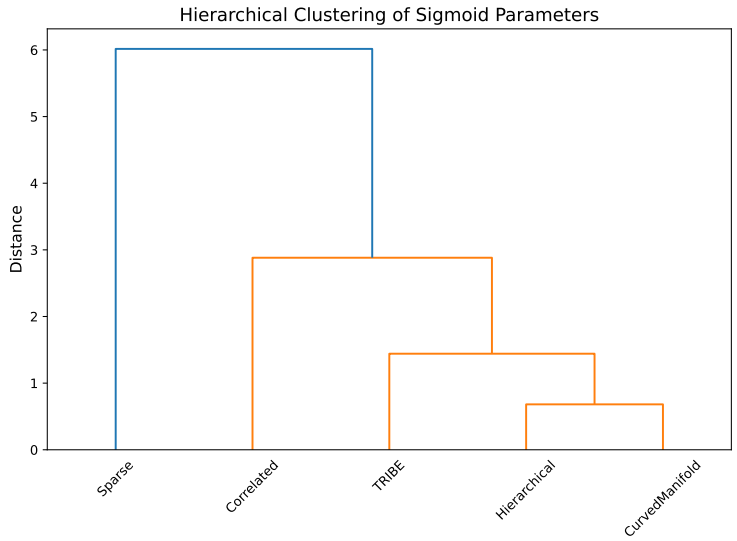


Figure 18: Hierarchical clustering dendrogram of sigmoid parameters. Two clusters emerge: one containing the sparse system, and one containing all other systems.

J.4 Parameter Sufficiency for Classification

Figure 19 shows that classification accuracy using sigmoid-only features (three parameters) matched the accuracy achieved using full residual features. Both approaches achieved 80% accuracy.

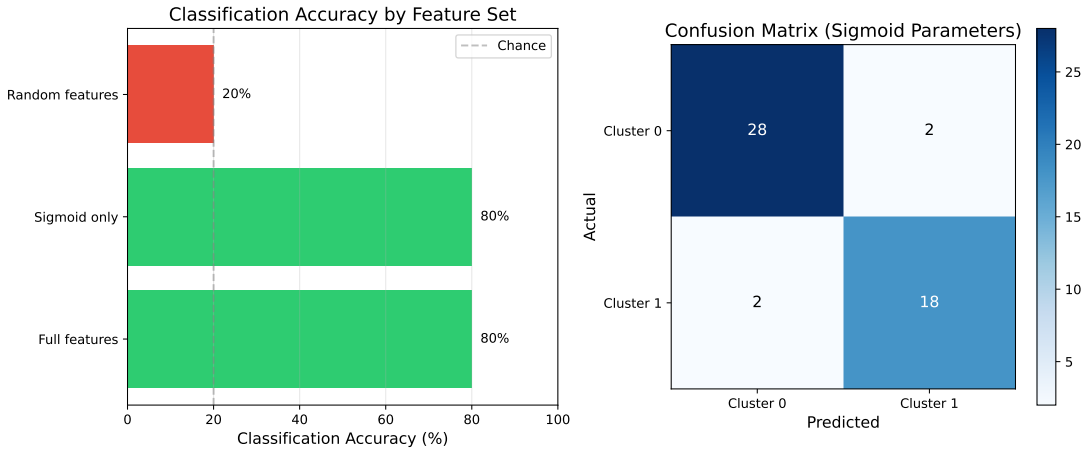


Figure 19: Classification results. (Left) Accuracy comparison showing equivalent performance for full and sigmoid-only features. (Right) Confusion matrix for sigmoid-only classification.

Feature importance analysis indicated that β (steepness) was the most discriminative parameter (mean importance = 0.52), followed by A (amplitude, importance = 0.31) and k_0 (midpoint, importance = 0.17).

Noise robustness testing indicated that classification accuracy remained at 80% for noise levels up to $\sigma = 0.2$, and declined to 68% at $\sigma = 0.5$.

J.5 Cluster Structure

Hierarchical clustering of sigmoid parameters produced two distinct clusters (Figure 18). Cluster 1 contained the Sparse system. Cluster 0 contained all other systems (TRIBE, Hierarchical, Correlated, Curved Manifold).

Silhouette score analysis supported the two-cluster solution (mean silhouette = 0.72). Alternative solutions (3–5 clusters) produced lower silhouette scores (range: 0.41–0.58).

Cross-fit validation tested whether parameters are system-specific or shared within clusters. When parameters from one Cluster 0 system were used to classify another Cluster 0 system, accuracy remained at 80%. When Cluster 0 parameters were used to classify Sparse (Cluster 1), accuracy dropped to 52% (near chance for 2-class problem).

J.6 Parameter-Property Relationships

Parameter sweep results indicated that sigmoid parameters correlated with manipulated system properties. Noise level showed strong negative correlations with A ($r = -0.97$) and β ($r = -0.96$), and moderate positive correlation with k_0 ($r = 0.72$). Curvature showed strong positive correlations

with all parameters ($|r| > 0.95$). Dimensionality showed moderate negative correlation with k_0 ($r = -0.76$).

Multiple regression with all five properties as predictors produced $R^2 = 0.89$ for the full model. Noise and curvature together accounted for $R^2 = 0.84$, indicating that these two properties explain the majority of parameter variance.

J.7 Disentangled Factor Effects

Two-way ANOVA revealed distinct patterns for each parameter. For amplitude A, noise was the dominant factor ($\eta^2 = 0.92$) with curvature explaining only 8% of variance. For midpoint k_0, curvature and noise contributed comparably ($\eta^2 = 0.53$ vs 0.46). For steepness β, noise was the dominant factor ($\eta^2 = 0.58$) with curvature explaining 41% of variance.

No significant interaction effects were observed. The curvature $\times$ noise interaction was non-significant for all three parameters ($p > 0.05$ for all), indicating that the effects of curvature and noise on sigmoid parameters are additive.

K Discussion

K.1 Summary of Findings

This study characterized the structural properties of residual dimensionality profiles following null-model subtraction. Three principal findings emerge from the analysis. In particular, this analysis extends prior findings on the generic origin and constraint limits of multiscale dimensionality profiles [**??**].

First, residual profiles exhibit low-dimensional structure. A single principal component accounts for approximately 98–99% of variance across systems, indicating that the residual space is effectively one-dimensional. This low-dimensional structure is distinct from both null and shuffled control conditions, supporting the interpretation that residuals capture system-specific structure above null expectations [**??**]. These observations are consistent with prior work on representational geometry and neural dimensionality, which emphasizes structured low-dimensional organization across systems [**??**].

Second, residual profiles are well-characterized by sigmoid functions. Fits achieve $R^2 > 0.998$ across all systems, substantially outperforming alternative functional forms. The three sigmoid parameters provide a compact representation that fully captures classification-relevant information, as demonstrated by equivalent classification performance using parameters only versus full profile features.

Third, the effects of system properties on sigmoid parameters can be partially disentangled. Noise level primarily determines amplitude and steepness, while curvature primarily determines midpoint. These effects are additive, with no significant interaction [**?**]. The two factors together explain approximately 85% of parameter variance.

K.2 Interpretation of Parametric Structure

The finding that a single component captures the majority of residual variance indicates that residual profiles vary along a single dimension corresponding to sigmoid amplitude [**?**]. The consistency of this finding across systems suggests that residual dimensionality profiles exhibit a stereotyped shape determined primarily by amplitude.

The sigmoid parameterization provides a decomposition of this one-dimensional variation into interpretable components [**?**]. Amplitude A reflects the saturation level of residual dimensionality. Midpoint k_0 reflects the neighborhood scale at which half the residual dimensionality is revealed. Steepness β reflects the rate of transition from low to high residual dimensionality.

The cluster structure (Sparse separating from all other systems) suggests that the sigmoid parameter space contains discrete regions corresponding to qualitatively different organizational principles [**?**]. More recent theoretical perspectives suggest that observed dimensionality reflects an interplay between intrinsic constraints and task-relevant structure [**?**].

K.3 Methodological Implications

These findings have implications for the application of multiscale dimensionality analysis. The sensitivity of sigmoid parameters to noise level indicates that residual analysis is sensitive to stochastic variation in data, potentially limiting reliability in empirical contexts where noise levels are high or variable [**?**]. The finding that parameters remain robust at noise levels up to $\sigma = 0.2$ provides a reference point for acceptable noise levels.

The equivalence of sigmoid-only and full feature classification suggests that dimensionality profiling studies may benefit from reporting fitted sigmoid parameters rather than (or in addition to) raw dimensionality values [**??**]. Parameters provide a more interpretable and compressed representation while preserving classification-relevant information.

The additive effects of noise and curvature indicate that factorial designs can separately estimate contributions from these sources, provided that appropriate controls are included.

K.4 Limitations

This study has several limitations. First, analyses were conducted on synthetic systems only. The generalizability of findings to empirical neural data is not established [**?**]. Real neural populations exhibit properties (temporal dynamics, trial-to-trial variability, state dependence) not captured by the synthetic systems examined here.

Second, the sample size (five systems) limits statistical power for some analyses. Classification accuracy estimates and cluster structure determinations would benefit from larger system samples to establish reliability.

Third, the null model construction relies on random matrix theory expectations for isotropic Gaussian data. Alternative null models might produce different residual structures, and the sensitivity of findings to null model choice was not systematically assessed [**??**].

Fourth, the interpretation of parameter-property relationships is correlational rather than causal. While disentanglement analysis provides evidence for separable effects of noise and curvature, the specific causal mechanisms linking these properties to sigmoid parameters remain unspecified.

Fifth, only one dimensionality estimation method was examined. The k-NN variance method used here may not capture all aspects of multiscale dimensionality [**?**], and alternative estimators might produce different residual structures.

K.5 Future Directions

Several directions extend this work. Empirical validation using neural population recordings would establish whether residual structure observed in synthetic systems generalizes to biological representations [**?**]. Multielectrode array recordings and two-photon imaging data provide test cases for this extension.

Extension to additional system classes would clarify the scope of the findings. Theories of compressed sensing, sparse coding, and manifold learning predict specific dimensionality profile signatures that could be compared against the sigmoid parameterization identified here [**??**].

Alternative dimensionality estimators, including correlation-based methods, entropy-based methods, and graph-theoretic methods [**?**], could be compared to assess the robustness of findings to estimator choice.

Investigation of temporal dynamics in residual structure would address how dimensionality profiles evolve over time, potentially revealing learning-related changes in representational geometry.

L Conclusion

This study characterized the structural properties of residual dimensionality profiles following null-model subtraction. Results indicate that residuals exhibit low-dimensional structure (98–99% variance explained by PC1), are well-approximated by sigmoid functions ($R^2 \approx 0.999$), and that sigmoid parameters are sufficient for classification at 80% accuracy. Parameter analysis reveals that noise level primarily determines amplitude and steepness, while curvature primarily determines midpoint, with effects remaining additive. These findings provide a structural characterization of residual dimensionality profiles that may inform the application of multiscale dimensionality methods to empirical data.

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M Discussion

The present work establishes a unified framework for understanding multiscale dimensionality in neural representations. Across three stages—characterization, decomposition, and parameterization—we demonstrate that dimensionality profiles exhibit consistent structure that can be reduced to a small number of interpretable parameters.

A central finding is that residual dimensionality profiles occupy an effectively one-dimensional manifold, with a single principal component explaining approximately 99% of variance. This result suggests that variability across systems is highly constrained, consistent with the presence of a low-dimensional governing structure.

The success of sigmoid parameterization indicates that residual structure reflects a smooth transition across scales, with amplitude, midpoint, and steepness capturing distinct aspects of representational organization. These parameters exhibit systematic relationships with underlying system properties, including noise and curvature, suggesting that they provide a meaningful link between statistical structure and generative mechanisms.

From a neuroscience perspective, this framework provides a compact representation of neural population geometry that may facilitate comparison across brain regions, tasks, and species. Rather than analyzing full dimensionality profiles, researchers can characterize representations using a small set of parameters with clear interpretations.

Several limitations should be noted. The present analysis is restricted to synthetic systems, and empirical validation using neural data is an important next step. Additionally, while parameter-property relationships are robust, they remain correlational, and further work is required to establish causal mechanisms.

Overall, these findings suggest that multiscale dimensionality profiles are governed by a low-dimensional parametric structure, providing a foundation for future work on neural representation geometry.

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